

Pre-service Teacher Perceptions of ChatGPT for Lesson Plan Generation

Journal of Educational Technology
Systems
2025, Vol. 53(3) 219–241
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DOI: 10.1177/00472395241301388
journals.sagepub.com/home/ets



Peter J. Kalenda¹ , Logan Rath²,
Mariam Abugasea Heidt² ,
and Allison Wright²

Abstract

This study explores preservice education student perceptions on ChatGPT's ability to write lesson plans through the use of a guided analysis process with 59 undergraduate and graduate students enrolled in STEM, TESOL, and Social studies methods courses for grades PreK–12. A pre- post-survey and a reflection on the strengths and weaknesses of ChatGPT were written by participants. Statistically significant findings were revealed regarding student perceptions of ChatGPT's ability to write a complete lesson plan in their content area and ChatGPT's ability to write a detailed lesson plan for a substitute teacher. In both cases, students' perceptions of ChatGPT to complete these tasks decreased after they engaged in analysis of the content. Students shared that they plan to use ChatGPT for lesson planning as a practicing educator, but indicated that revision of output is necessary. A call for future work is explored.

Keywords

artificial intelligence, generative AI, lesson planning, education, survey, curriculum, analysis

¹SUNY Geneseo, Geneseo, NY, USA

²SUNY Brockport, Brockport, NY, USA

Corresponding Author:

Peter J. Kalenda, SUNY Geneseo, 1 College Circle, Geneseo, NY 14454, USA.

Email: pkalenda@geneseo.edu

Introduction

ChatGPT, a generative artificial intelligence (AI) tool that provides lengthy responses to natural language questions, became popular on our campuses in the spring of 2023. With the diffusion of this new tool, we began to hear horror stories from our colleagues of students using this tool to complete their assignments as well as the ways in which the tool failed spectacularly. As university-based teacher educators, we felt it was our duty to understand how this tool could be used effectively in K-12 settings. Through discussions with trusted in-service teachers, we learned that some school districts in our area were embracing this tool and encouraging teachers to use generative AI to develop curriculum including student learning outcomes and even write entire lesson plans. These teachers shared that the bulk of the work was done for them by the generative AI tool and they just needed to do some light editing to make the lessons work for their students.

In fall of 2023, two authors, a professor and a non-professor researcher, were discussing the fears and concerns of their colleagues on campus and decided to investigate ChatGPT's ability to generate a lesson plan as if they were completing an assignment for the professor's course. The non-professor researcher entered the details of the assignment and asked ChatGPT to generate a lesson plan as if it were to be submitted. The results were surprising to the pair; but upon deeper analysis, it became clear that the lesson plan would need significant editing. The two then decided to plan a pilot study to determine if preservice childhood math and science methods students would come to similar conclusions. This design included a guided analysis process to determine the strengths and weaknesses of the generated lesson plan with a pretest and posttest and written reflection for data collection. The outcomes fascinated the authors and they were eager to expand their study to include other content areas in education. The professor and non-professor researcher then reached out to colleagues in two new content areas: social studies and Teaching English to Speakers of Other Languages, or TESOL. As a combined group, the present study was born with the following research question: How do students' preconceptions of ChatGPT's ability to generate a lesson plan compare to their analysis of ChatGPT generated lesson plans? The research team for this study is composed of three professors and one non-professor researcher.

Literature Review

Generative Artificial Intelligence (AI) is a technology that uses deep learning models to generate human-like content in response to varied and complex prompts, and ChatGPT (Chat Generative Pre-Trained Transformer) is a specific AI tool that is driven by a chatbot (Michel-Villarreal et al., 2023). The primary focus of generative AI tools was originally to mimic human conversation and intelligence; however, this has rapidly expanded to higher-level tasks, such as creating poetry, coding, and writing

essays (Yusuf et al., 2024). The rapid evolution of generative AI models can be attributed to a convergence of factors including access to large data sets, cloud computing, data harvesting, storage, advances in algorithms, and increases in computational power (Kurtz et al., 2024; Walczak & Cellary, 2023). General access to generative AI tools did exist before ChatGPT's launch in November 2022, specifically, dating back to the 1940s with early work on neural networks simulating biological neurons (AlAli et al., 2024). However, ChatGPT has quickly become the fastest growing generative AI tool attracting wide interest and discussion in the education world because it is open access (Nikolopoulou, 2024). This free public access has leveled the playing field across educational settings (van den Berg & du Plessis, 2023). Furthermore, recent graduates of education programs are reporting that they would use ChatGPT for lesson planning, so it is critical that we begin to prepare our pre-service teachers to utilize generative AI tools responsibly (Corp & Revelle, 2023).

The introduction of ChatGPT in 2022 has led to its application in many educational contexts, including lesson planning (Mintz et al., 2023). Teachers can utilize a variety of generative AI tools other than ChatGPT for lesson plan creation, such as, LessonPlans.ai, MagicSchool.ai, Google Gemini, and Microsoft Copilot. With the wide availability of such tools, researchers have started to question the strengths and weaknesses of generative AI tools to complete complex tasks like lesson planning, which has led to the publication of many research studies in recent years. An overview of generative AI lesson planning research from 2023 and 2024 is explored below.

Generative AI tools can be a useful support for lesson planning, however, college faculty need to be careful working with pre-service teachers because they need to understand the value of the tool before they adopt it (Baytak, 2024; Powell & Courchesne, 2024). ChatGPT often has problematic lesson plan components, such as missing details and inaccurate information. It is important to develop a skeptical disposition regarding AI outputs and also determine why, how, and when to best use generative AI tools with their future students. Pre-service teachers will need help to learn how to thoroughly examine AI outputs, which can best be done through a scaffolded approach. Furthermore, they will need additional assistance learning how to properly revise their generative AI lesson plan to align with local learning standards. Studies of students using generative AI tools in college and university settings have shown that they do begin to recognize concerns regarding accuracy, privacy, ethical issues, and the impact on their own professional development (Chan & Hu, 2023).

Lesson plans may not initially align with teachers' preferences and iterative refinement may be necessary to reach the desired goal (Koraishi, 2023). Furthermore, ChatGPT lesson plans may sometimes provide vague references to items like videos or warm-up activities which may also require elaboration through further prompts provided to the generative AI tool. When asked for elaboration, ChatGPT often provides guidance to video resources with insight on how to use these resources. This iterative collaborative approach to lesson planning between ChatGPT and teachers leads to a

new approach to lesson planning and the creation of other instructional materials we were not able to utilize just a few years ago. It is important to note that ChatGPT alone is not a perfect system for the creation of lesson plans, but it is rather a helpful tool or assistant in the lesson plan creation process. An actual human teacher will be needed for the success in teaching, learning, and assessment (Kehoe, 2023).

When using ChatGPT to develop high school chemistry lessons, ChatGPT has been found to have many strengths regarding creating lesson plan outlines, recommending resources, using accurate content knowledge, discussing instructional strategies, differentiating content for different learning levels of students, and generating assessment questions (Clark et al., 2024). The tool has limitations regarding the ability to create visual aids, slides, or specific complete lesson plans. ChatGPT is a helpful supplement, but can not replace the role of the teacher with creating lesson plans and additional instructional materials. On the contrary, studies conducted in Korea on the ability of pre-service teachers to utilize ChatGPT for the creation of science lesson plans showed that students were already critical of content created by generative AI and were moderately successfully in developing lesson plans with this software without much support needed from faculty (Lee & Zhai, 2024).

Broutin (2024) studied teacher candidate's use of ChatGPT to lesson plan for math and discovered that teacher candidates use this tool for tutoring (asking ChatGPT questions), asking for guidance, validation of lesson plan choices, simulating if a lesson plan choice would work, organizing or sequencing lesson plans, and adjusting lesson plans through the use of interactive refinement. Additionally, a study conducted by Karaman (2024) showed that primary students who were taught math lesson plans created by ChatGPT had their academic achievement significantly increase when compared with a control group using the current school's math curriculum.

Studies regarding special education teachers utilizing generative AI for lesson planning also show that these tools can significantly enhance the lesson planning process (Khazanchi & Khazanchi, 2024). Generative AI can also help level the playing field for students with disabilities and international students, providing many assistive tools that could significantly help them succeed (Farrelly & Baker, 2023). Specifically, using a variety of generative AI tools can lead to the production of diverse content, including images, texts, and videos, which has the potential to revolutionize the teaching and learning experience for these students. These generative AI tools do however lack a human component, so it is important to still consider empathy, creativity, and unique student needs when using them as an assistant during the lesson planning process.

Teachers who have used ChatGPT for lesson planning have shared that this tool is useful for assisting with teaching and learning aspects of lesson planning, such as enhancing learning outcomes, increasing student engagement and motivation, and improving teaching practices and pedagogical beliefs (ElSayary, 2023). However, the lesson plans were weak in the areas of assessment and feedback. The biggest challenges teachers reported were poor information accuracy and bias. DeCarlo et al.

(2024) suggests that pre-service teachers can and should utilize generative AI to assist with analyzing relevant student assessment data. Transition, Sakamoto et al. (2024) noted that lesson plans created with ChatGPT without sufficient student data were not adequate in their design. Building these data literacy skills with generative AI tools will allow pre-service teachers to use data for more effective and targeted classroom instruction. However, there are drawbacks and ethical concerns to consider, such as handling of sensitive student data (Khazanchi & Khazanchi, 2024; Yeralan & Lee, 2023). Educators must ensure the security and privacy of student data to protect student confidentiality and uphold state and federal privacy regulations (Khalid & Ahmad, 2024).

The potential of ChatGPT is exciting and provides many new opportunities for pedagogical innovation and creativity (Hodge-Zickerman & York, 2024); however, these tools should be used with caution and we need to critically evaluate their limitations and biases when they are used to support teaching and learning (Mosaiyebzadeh et al., 2023; van den Berg & du Plessis, 2023). Pre-service and student teachers need to be equipped with the skills necessary to critically analyze instructional materials (Cun & Huang, 2024). Specifically, they need to be able to determine what is accurate and pedagogically sound (Corp & Revelle, 2023). This can be accomplished by practicing with ChatGPT in two ways: (1) Writing and revising prompts to more effectively create lesson plans that align with students' needs, interests, and cultural backgrounds, and (2) practice evaluating lessons created by ChatGPT. They also need to know how to use these tools ethically and need access to an AI literacy guide to support their work (Baytak, 2024; Kehoe, 2023). Once pre-service teachers gain these skills and experience with using generative AI tools they increase their use of interactive AI-powered tools in learning scenarios with their own students (Uzumcu & Acilmis, 2024). However, it is important to note that training teachers to do these skills is a complex task and teacher educators require their own professional learning to be able to train their students (Dotan et al., 2024; Malik & Aslam, 2024; Saimon et al., 2024).

It is also important to keep in mind learner preferences regarding generative AI use. A study by Smolansky et al. (2023) revealed that some higher education students have reservations with using generative AI as it can be seen as stifling creativity. The authors suggest that less focus should be placed on teaching students generative AI skills, and rather assist students with navigating the more complex interplay between technology, cognition, social interactions and value. Conversely, Chiu (2024) found that university students do want to learn how to use generative AI tools in order to improve their future job perspectives, as employers are starting to look for generative AI skills among their applicants (Hajam & Gahir, 2024; Hashmi & Bal, 2024).

Ultimately, it is the university's obligation to prepare their students for the ever-evolving demands of industry, which appear to be currently focused on AI-related skills (Lee et al., 2024). It is essential that efforts are made to cultivate positive attitudes toward generative AI tools among pre-service teachers to further the development of

innovative pedagogies (Adeyele & Ramnarain, 2024). Focus should also be placed on developing the whole person, with an emphasis on personality traits, such as perseverance and grit, so our students have the ability to adapt and persist in an ever-changing world (Dai et al., 2023).

Contexts

This study occurred in three different educational contexts during the Spring 2024 semester at two different comprehensive universities in the Northeast United States. All three contexts were teacher education programs, however, the content-focus in each scenario varied.

Undergraduate-Level Childhood STEM Education

Participants were recruited from two sections of an undergraduate, in-person Childhood STEM Education methods course. Students were all sophomores or juniors who were seeking teaching certification in early childhood, childhood, or childhood with special education. These students were in their second semester of education coursework and have only written one lesson plan before enrolling in this course. A total of 46 students were enrolled in these sections and 38 students opted to provide data to this study.

Undergraduate Childhood Social Studies

Participants included students from two sections of an undergraduate hybrid social studies methods course required for completion of their initial teacher certification in a Childhood Inclusive preparation program. Students were juniors and seniors in the second phase of a four-phase program. A total of 33 students were enrolled in the two sections of the course with 12 students opting to take part in the study.

Graduate-Level TESOL and Bilingual Education in-Service Teachers

Participants were recruited from two sections of a fully online, asynchronous methods course that was designed for both TESOL and Bilingual Education students, but primarily focused on TESOL methods. Students in these programs are graduate students and are already certified teachers. They are completing the TESOL and/or Bilingual Education programs (MSEd and Advanced Certificate levels available) as a requirement for continued certification or/and to add on a new certification or certifications. The course is their second methods course, assuming their familiarity with lesson planning for emergent bilinguals. There were 38 students (henceforth being referred to as TESOL students for simplicity, as many Bilingual Education students are also gaining TESOL certification) between the two sections, with nine students choosing to participate in the study.

Course Assignment Commonalities

Though this study was conducted in different content areas in the field of teacher education, there were commonalities between all of the course assignments for this project. For the assignments, the researchers used ChatGPT (v3.5) to generate content-specific lesson plans, each with items specific to their field (the inquiry-based BCBS 5E Model for STEM, different topics for social studies, and language objectives for TESOL). These lesson plans were then shared with students in the courses to use in completing the assignment. The assignment required that the students critically analyze the AI-generated lesson plan(s) through a scaffolded process using content and pedagogical skills specific to their content areas, making note of how well the ChatGPT-generated plans conformed (or did not conform) to what might constitute strong lesson plans in their discipline (Kalenda, 2024). Lastly, after critical analysis, students summarized their findings in narrative format, noting the strengths and weaknesses of AI-generated lesson plans.

Methods

This study used a pretest-posttest design to measure the participants' change in perception of the quality of generative AI lesson plans. A pretest-posttest design was chosen for its ability to measure change surrounding an intervention (Creswell & Creswell, 2023). Upon obtaining ethics approval for the study by the lead PI's institutional review board, the non-professor researcher became the point person for liaising between each of the different classrooms where the study was taking place. The study consisted of three steps in each context: a pretest where students predicted how well ChatGPT would develop a lesson plan that met specific criteria, an educational intervention where students analyzed a lesson generated by ChatGPT through a scaffolded process, and a posttest and reflection where students evaluated how well ChatGPT accomplished the task.

Participants

Participants in this study were undergraduate and graduate students enrolled in methods courses as part of a teacher certification program at a large public university system in the northeast United States. All participants were over the age of 18 and no further demographic information was collected. Three different methods courses were chosen for this study: childhood STEM methods, childhood social studies methods, and P-12 Teaching English to Speakers of Other Languages (TESOL) methods.

Data Collection

The professor for each course explained the entire project and then distributed the informed consent. Either a link to the survey (social studies and TESOL groups) or

a paper copy of the survey (STEM group) was provided. For the fully online TESOL course, the professor and another researcher recorded a video describing the project and encouraging participation while explaining the extent to which responses would be anonymous. In order to track pretest and posttest responses, participants received a random number. In-person classes (STEM and social studies) received their number on a randomly distributed index card and were encouraged to store the card in a safe place and take a photo of the card with their phone. Students in the online class received a random number generated by Qualtrics and checked a box to confirm they had recorded the number somewhere for later use. Between the pretest and posttest surveys, students completed a regularly scheduled lesson planning activity. In previous semesters, students either created a lesson from scratch or critiqued a lesson from a published curriculum. For this study, students received a lesson plan generated by ChatGPT and completed a scaffolded analysis following guided prompts from the professor.

Instruments

The instruments for this study involved a pretest, a posttest, and a lesson plan created by ChatGPT version 3.5.

Pretest and Posttest. The pretest and posttest were created by the non-professor researcher using Qualtrics (<https://www.qualtrics.com>) and anonymous links were shared with the professor to share with their class. A paper version of each instrument was also created and was distributed in the childhood STEM group. To ensure consistency between the pretest and posttest, the individual questions were worded exactly the same in each course. The introductory text of the question set signified the difference between the instruments. The pretest was labeled “this set of questions asks how well do you think ChatGPT could assist with writing the following aspects of a lesson plan,” whereas the posttest was labeled “this set of questions asks how well you think ChatGPT did assist with writing the following aspects of a lesson plan.” Questions were closed-ended and used five-point Likert-scale questions with responses ranging from “Very Good” to “Very Poor.” The specific constructs measured were the ability of ChatGPT to:

1. Write a complete lesson plan
2. Align with a specific performance standard
3. Provide scaffolds to support learning of performance standards
4. Include a final performance assessment
5. Include references of learning theories that were used
6. Differentiate approaches to learning
7. Provide detailed enough plans for a substitute teacher

Across all settings, the pretest and posttest measured equal constructs, but the specific language of the questions for participants were customized to the course content. When measuring alignment with specific performance standards, the STEM question stated: “Align a STEM lesson plan with both a NYS math and science learning standard.” The social studies question stated: “Align a social studies lesson plan with the NCSS C3 Framework.” The TESOL question stated: “Align to appropriate national and NYS standards.” Whereas each question was customized to its unique content area, all three questions measure the perceptions of ChatGPT’s ability to align with an appropriate performance standard. The final question of the posttest asked participants to upload a written reflection on their analysis of the lesson plan. For the STEM and social studies, pre-service teachers were asked to write a one-page letter to their future self about the strengths and weaknesses of using ChatGPT for writing lesson plans. For the TESOL group, students were asked to answer the following question: “In looking at the AI-generated lesson plan, what are some important takeaways you had and some implications you think this has for teaching and learning?”

Lesson Plan Creation and Evaluation. The educational intervention in each group varied in each group. All interventions shared the overarching goal of evaluating an AI-generated lesson plan that was unique to the context of the course. In the STEM group, the instructor created a list of grade-specific paired math and science standards so that there would be between two and three unique lesson plans per grade level. The non-professor researcher then used the paired prompts to create each lesson. The responses from ChatGPT were not modified, but were copied into individual Google docs that were assigned a grade level and number (e.g., KA, KB, KC for Kindergarten; 1A, 1B, 1C for 1st grade, etc.). Lessons were checked for formatting consistency and were saved in a Google drive folder for student access. An example prompt for lesson plan 2B was as follows:

Create a STEM lesson plan for 2nd grade using the 5E model that covers 3 days of instruction and uses a children’s storybook to engage learners. The lesson should align with this science standard: Make observations of plants and animals to compare the diversity of life in different habitats. The lesson should also align with this math standard: Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare problems using information presented in a picture graph or a bar graph. Include a final performance assessment that is aligned with both learning standards. Include 3 references of learning theories that were used in the lesson plan. Ensure that the lesson plan is designed using principles from Universal Design for Learning.

Twenty lesson plans in total were created and shared with students. Students were enrolled in various grade level focused programs (early childhood, childhood, childhood with special education), so participants worked in self-selected groups of two

to select a grade level lesson plan of their choosing that best fit their planned certification pathway.

In the social studies group, the professor created one lesson plan at each grade level for the topic chosen by that group at the start of the semester. The prompts were then entered into ChatGPT and the unmodified responses were printed and distributed in person to each group. One specific prompt for a 5th grade lesson plan focusing on Marginalized Groups' Perspective of American History was:

Create a social studies lesson plan for 5th grade focusing on marginalized groups' perspectives of American History. The lesson plan should connect to the 5 threads of social studies (political, geographic, global, economic, sociocultural). The lesson should align with the NCSS 3C framework. Include 3 references of learning theories that were used in the lesson plan. Ensure that the lesson plan is designed using principles from Universal Design for Learning.

In pairs, one specific lesson plan based on the students' specific topics were chosen for analysis. The students were provided time in class to begin their initial review of the lesson plan using prompts provided in the assignment outline. They were also given time during the next class to continue to work together to complete their narratives based on their analysis as well as complete a table specifically targeting components needed in a comprehensive lesson plan as a way to create a side-by-side comparison with the ChatGPT-generated lesson plan.

In the TESOL group, the non-professor researcher and professor co-created one lesson plan that was used for analysis by each student individually. The content of the lesson was a second grade science lesson on plant growth and seed reproduction that was meant to support emergent bilinguals. The prompt was entered into ChatGPT using the Next Generation Science Standards (NGSS) abbreviations in lieu of the language of the topics to simulate communication between a mainstream and English to speakers of other languages (ESOL) teacher. The specific prompt for the lesson plan was:

Build me a lesson plan for English Language Learners for Science using NGSS standards: 2-LS2-1 and 2-LS2-2. The lesson plan needs goals, Big Ideas, Essential Questions, Content Objectives, Language Objectives, a performance assessment that has a task and lists all evidence to be collected. The lesson should also build background knowledge by connecting to students' prior knowledge. Include a table with time for each activity, the name of the activity, differentiation for different levels of learners, and a list of resources for each activity. Also list any key vocabulary terms and at the end list all the supporting materials and resources necessary. Provide several strategies that can be used to accommodate diverse learners.

The lesson plan was then posted in the learning management system along with a rubric for the students to analyze the contents of the lesson plan and write a reflective

statement. The rubric students used to assess the lesson plan was the same rubric the professor uses to score students on the creation of their own lesson plans for emergent bilinguals in the first methods class. Students also had access to the lesson plan template for the program and were told to score the AI-generated lesson plan using the rubric, with an explanation after each score, and then include a conclusion explaining important takeaways they had from the exercise of analyzing an AI-generated lesson plan as well as implications for teaching and learning.

Data Analysis

Completed survey data was downloaded from Qualtrics into Microsoft Excel. Qualtrics's recode values setting was used to assign a value of 5 to "Very Good" and 1 to "Very Poor." Variables across each data set were matched up based on the seven common constructs. For each group, pretest and posttest files were combined into one workbook and the XLOOKUP function in Excel was used to match respondents based on the randomly generated number they received. Responses that did not have values for the posttest variables were removed from the dataset so that only complete responses remained. Finally, all three groups were combined into one data set. Those data were then exported as a comma-separated file and imported into jamovi version 2.3 (The jamovi project, 2022) and ran a paired samples t-test with descriptive plots. A paired samples 2-tailed t-test was chosen to measure the mean differences in responses for each pretest-posttest pair (Creswell & Creswell, 2023). The output was then saved and shared with researchers for interpretation. Inductive approaches to qualitative data analysis (Saldaña, 2021) were used to analyze the responses to the open-ended question. Each set of responses had two reviewers, the course professor and the non-professor researcher. Intercooder agreement was reached by consensus in accordance with Saldaña's (2021) recommendation for team-based coding. Results were then discussed among all authors to ensure transferability.

Findings

Quantitative Survey Results

The analysis of combined survey data from all three classes ($n = 59$; 38 STEM, 12 social studies, 9 TESOL) allowed for the researchers to identify statistical trends among all three classes (see Table 1). Each of the seven shared constructs measured are explored below.

Ability to Write a Complete Lesson Plan. Findings were statistically significant, $t(58) = 2.039$, $p = 0.046$, regarding student perceptions of ChatGPT's ability to write a complete lesson plan in their content area (Figure 1). The data show that participants predicted that ChatGPT would perform well with writing a complete lesson plan, but their

Table 1. 2-Tailed t Test for Variables.

	Paired Differences					t	df	Sig.
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Write a complete lesson plan	0.254	0.958	0.125	0.005	0.504	2.039	58	0.046
Align with a specific performance standard	−0.153	1.186	0.154	−0.462	0.157	−0.988	58	0.327
Provide scaffolds to support learning of performance standards	0.051	1.057	0.138	−0.225	0.326	0.369	58	0.713
Include a final performance assessment	−0.085	1.25	0.163	−0.41	0.241	−0.521	58	0.604
Include 3 references of learning theories that were used	−0.237	1.466	0.191	−0.619	0.145	−1.243	58	0.219
Differentiate approaches to learning	−0.305	1.405	0.183	−0.671	0.061	−1.668	58	0.101
Provide detailed enough plans for a substitute teacher	0.966	1.245	0.162	0.642	1.291	5.959	58	<.001

post-assessment revealed a significant decrease in the perceived performance of ChatGPT to complete this task.

Provide Detailed Enough Plans for a Substitute Teacher. Average participant assessment of ChatGPT’s ability to write a detailed enough lesson plan for a substitute teacher dropped between the pre- and post-assessment (Figure 2). The findings are statistically significant, $t(58)=5.959$, $p<.001$, that engaging in our scaffolded analysis process led participants to determine that ChatGPT lessons produced through one original prompt were not detailed enough for substitute use as generated.

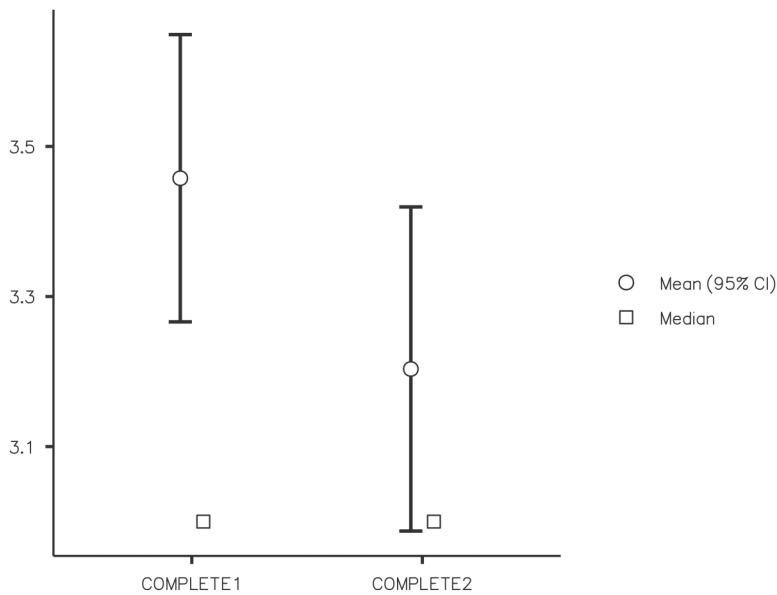


Figure 1. Write a complete lesson plan.

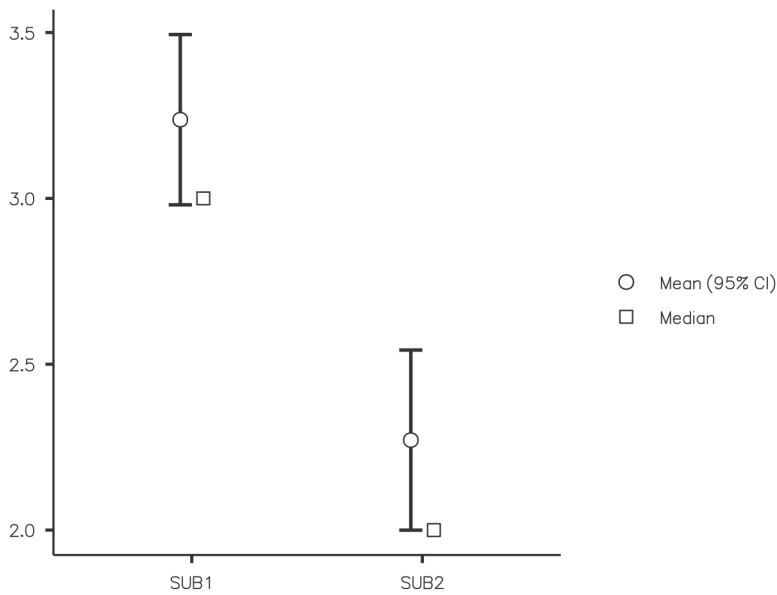


Figure 2. Provide detailed enough plans for a substitute teacher.

Additional Shared Constructs. The analysis of the five remaining shared constructs were not statistically significant, but did reveal general trends that were positive or negative. Student perception regarding the ability of ChatGPT to provide scaffolding throughout the lesson plan decreased after the lesson plan analysis. However, the other four remaining items showed an increase in student perceived performance of ChatGPT after the analysis. These four areas include the ability of ChatGPT lessons to align with a performance standard, include a performance assessment, include three references on learning theories used in the lesson plan, and differentiate instruction to meet diverse learning needs.

Childhood STEM Education Results

Qualitative analysis of student reflective letters ($n = 32$) showed that students would utilize ChatGPT for lesson planning in the future, but with some reservations related to designing a complete lesson plan, writing a detailed lesson plan for a substitute, properly aligning with a learning objective/standard, containing inaccurate STEM information (hallucinations), or utilizing universal design for learning (UDL) to meet the unique learning needs of students. Students shared that they would start lesson planning with ChatGPT, but would need to complete revisions before the product was ready for use. The following student quote captured shared student sentiments:

“We have learned that generative AI can be a successful tool in creating outlines of a lesson plan... . In contrast, however, many times these lesson plans are incomplete or contain information that can be fabricated by the AI itself. With this, we should use this tool to help in our planning endeavors but always make sure to add personalized data after this outline is completed to ensure that all students meet the learning objectives and can have an equal opportunity to do so.”

Conversely, students were consistently pleased with ChatGPT’s ability to write lesson plans utilizing the BSCS 5E model to support inquiry-based approaches to teaching and learning. They shared that the generative AI tool knew each of the 5E steps, used each step properly, and also raised helpful questions to raise curiosity among students.

TESOL Education Results

Of the 38 students in the two TESOL sections, nine students contributed to the quantitative data from the pre- and posttests and seven of those students completed the qualitative portion where they were asked to share their takeaways and the pros and cons of the AI-generated lesson plan. Though the numbers are modest, both the quantitative

and qualitative data from the TESOL course participants showed that they thought that the AI-generated lesson plans were incomplete and had insufficient detail for a substitute teacher to pick up and use. In both cases, participants had thought that the AI-generated lesson plans would do better on these metrics than they did. In the end, they saw AI-generated lesson plans as a “starting point” and place to get ideas. But ultimately, they “believe that teachers should not just rely on AI generated lesson plans as they are not adequate enough to have a great lesson on its own.” This points to the utility of AI-generated lessons, but with the reaffirming of a common caveat among scholars that a human element and critical thinking skills are needed to get the most out of AI (e.g., Chan & Hu, 2023; Cornish & Larter, 2024; Damiano et al., 2024; Onal & Kulavuz-Onal, 2024).

Childhood Social Studies Education Results

When looked at independently, the quantitative results within the social studies methods courses were different from the STEM and TESOL methods courses. The pre-survey responses from the students in the social studies methods classes indicated the students had *low expectations* that ChatGPT would be successful generating lesson plans based on their respective topics, and their perceptions of the assessment of the lesson plans that ChatGPT did generate *declined* even further according to the post survey data from the 12 participants taking part in the study. The quantitative results from this group alone are not determined to be statistically significant for this group. There may be myriad reasons the participants felt ChatGPT fell short when generating lesson plans for their review and potential use, however, the low number of student survey responses and the lack of qualitative narrative responses does not allow for any major claims yet in this content area.

Discussion/Conclusion

The two significant findings in our study were that the AI-generated lesson plans did worse than students anticipated at presenting complete lesson plans and these lesson plans did not go into enough detail for a substitute teacher to pick up and use. The incompleteness was in lacking necessary elements of strong lesson plans, such as how the lesson would incorporate learners’ “funds of knowledge” (their cultural assets; Moll et al., 1992) and how it would help build a learning community in the case of the TESOL lesson plan. Though there were many elements included that one might find in a typical lesson plan, the lesson plan did not include all elements that are expected in a lesson plan for a methods course (where students are expected to be explicit about all aspects of a lesson as they are still learning how to hone their teaching crafts).

Additionally, the AI-generated lesson plans were not strong in providing scaffolding for students’ learning through differentiation of instruction for various learning

needs and learning styles, as well as for linguistic diversity. The lesson plans were one-size-fits all and the level of detail was lacking, leading the user to need to fill in the blanks and gaps themselves. For instance, the procedure part of the lesson plans were simplistic and did not give enough detail about steps for activities, who would do what, how long the activities would take, or what one would need for the activities. Materials were also missing (e.g., worksheets, reading materials), meaning the user would need to find, make, or have AI generate the materials to use in the lesson.

Perhaps an interesting facet of the findings is the fact that students in the courses, before seeing the AI-generated lesson plans for this study, had a higher opinion of what AI could produce. So, in looking at the pre-test and post-test results, it was clear that students' opinions of AI's abilities fell after seeing the lesson plans it could create. This points to students having an originally higher opinion of what AI is capable of producing, perhaps through personal experience or through their beliefs about the capabilities of AI technology.

It is important to note that an outcome of the critical analysis students were asked to do was that it allowed students to not only look at the AI-generated lesson plans with a critical lens, but, in the process, it also allowed students to have a better grasp of what constitutes a strong lesson plan in their discipline. Without having to look more closely at the elements of strong lesson plans in their discipline, students would not have been able to be critical of the AI-generated lesson plans.

Throughout the study, the pre and post surveys were used for quantitative data analysis, and narratives in the form of letters were submitted by some (but not all) students participating in the study for qualitative analysis. This does not mean that opportunities to observe more anecdotal results were not present and able to provide additional insights into how the students approached their critical reviews of the ChatGPT generated lesson plans. Since the overall requirements were multifaceted it provided real time observations and discussions with participants as they engaged throughout the process of analyzing their respective lesson plans during face-to-face meetings (for the STEM and social studies courses), and through discussion posts for online TESOL forums. It was apparent, throughout these interactions, that the participants enjoyed the experience.

When presented with the AI generated lesson plans, the participants immediately engaged with the materials and began their initial reviews. Since AI is relatively new, generally speaking, and only beginning to find its way into educational settings, the students appeared curious to review the lesson plans generated for them by ChatGPT. They were actively involved with the materials provided for them and the discussions were targeted and meaningful. Since the study was enmeshed with established assignments, the students' submissions revealed their attention to detail and convictions when articulating their rationales for their analyses. While the assignment submissions themselves cannot be included for formal analysis, they did provide meaningful anecdotal opportunities to reflect on and are noteworthy for discussion and consideration.

Beyond appearing to have greater levels of enthusiasm when approaching the analyses, the students in all three content areas appeared to approach their critical reviews with greater conviction. Again, this was observed anecdotally, however, it was noteworthy in each course. This was not the first time an analysis of existing lesson plans had been a requirement expected in the courses, but typically the lesson plans provided were from “reliable” educational sources, such as experienced teachers or education publishers. When this has been the case, the analysis of the resources were not as vigorously critiqued. This may be a result of the fact that the students at the time felt more intimidated when identifying potential weaknesses in lesson plans that had been created by “professionals” who were perceived to be more knowledgeable and competent. When analyzing the ChatGPT-generated lesson plans, however, the participants in this study demonstrated greater levels of confidence throughout their critical analysis of the strengths and weaknesses of the lesson plans.

Limitations

Because generative AI is still relatively new, and not only to students but to those working in academia, large scale and longitudinal research is still in development. As such, our sample size and groups of participants are small and have limited diversity. We drew from pre-service and in-service teachers in childhood STEM education, social studies education, and TESOL. Other content areas are missing, and some of the samples were not large (i.e., TESOL), yet when looking at all of the data together, the corpus of data from our work as a whole does present promising glimpses into students’ perceptions of generative AI. More research needs to be done, including more participants and from additional programs (e.g., math, English/ English Language Arts).

Additionally, one of the authors, having had a number of students every semester using generative AI to complete assignments for them without permission, was reluctant to embrace the integration of AI into coursework beyond having students analyze AI-generated lesson plans. This limited our ability to go a step further and have all students work with ChatGPT or other tools to design their own AI-generated lesson plans by engineering the prompts. This has been done with one of the programs, but should be implemented in all of the programs to gather more data on students’ experiences with using generative AI to create stronger lesson plans than the ones they were presented with.

Future Work

Future iterations of this study will be conducted using newer versions of ChatGPT as they become available for free to the public. Since this study was conducted, ChatGPT has already released the next version, ChatGPT 4o, which the authors will utilize in their upcoming semesters. The authors will also make shifts to their data collection

methods to increase student participation, both quantitative and qualitative. Author 1, for example, made subtle improvements in how he established the assignment requirements that helped improve student participation. Participation was still optional, but Author 1 made it easier for students to feel empowered to readily submit their work for use in the study. While the data collected for this study was substantial and meaningful, greater participation in all content area classes will increase the usefulness of the results when identifying overarching themes and trends among data.

This study was presented to students with the assumption that ChatGPT's ability to create a lesson plan was based on its own base coding. However, we did not get into discussions with students about the role of prompt engineering. Our next iteration of this study will include opportunities for students to modify the original prompt created by professors or draft their own prompts to see whether they can improve ChatGPT's lesson plan generation to better align with intended outcomes. They can also use iterative refinement approaches by submitting multiple requests to ChatGPT. Goodman et al. (2024), found that one prompt was not realistic or reliably accurate for planning quality, inquiry-based classroom instruction. However, when using multiple prompts to refine lesson plan creation, the tool was useful in generating high-quality, open-ended questions to enhance the learning process. Preservice teachers will need to go through a scaffolded learning process to develop the skills necessary to use ChatGPT iteratively for inquiry-based lesson plan creation.

Participants will also be encouraged to try alternative generative AI websites. Each different generative AI chatbot is trained with a different data set, so it is possible to get different lesson plans using similar prompts (Baytak, 2024). Some alternative generative AI suggestions will include Google Gemini, Microsoft Co-Pilot, MagicSchool.ai, and lessonplans.ai. This will allow students to strengthen their skills using many different generative AI websites as a helpful assistant as they enter their careers as educators.

Consent to Participate

The Institutional Review Board at SUNY Geneseo approved a waiver of written consent and substituted participation in the survey for consent.

Consent for Publication

Participants were advised in the informed consent language that their responses would be used for potential publication.

Data Availability

Upon reasonable request, the authors will make deidentified data available for review without a license for reuse of the data.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical Considerations

The SUNY Geneseo Institutional Review Board approved our interviews (approval: 202324010) on October 16, 2023. Respondents gave consent through submitting the survey.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Peter J. Kalenda  <https://orcid.org/0009-0000-2482-1282>

Mariam Abugasea Heidt  <https://orcid.org/0000-0003-2672-8318>

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Author biography

Peter J. Kalenda is an assistant professor in Childhood STEM Education. His areas of research include technology integration in STEM methods courses and program evaluation of informal science education programs.

Logan Rath is a librarian. His area of research is the intersection of information literacy and literacy as a social practice, effective library instruction, and effective integration of technology into collegiate learning.

Marium Abugasea Heidt is an associate professor in TESOL Education where she is program coordinator for the TESOL and Bilingual Education programs.

Allison Wright is an instructor for Childhood Social Studies Education at SUNY Brockport.